F â€” Adopt vs Refactor vs Rewrite vs Net-New decision table

Revision: 1 **Last modified:** 2026-06-10T00:00:00Z

> Per-module disposition of the reference BobaLink implementation (44 `.ts` files under `docs/research/browser_extension/Browser Torrent Extension Guide/src/src/`) for the Boba browser-extension build. Decisions are evidence-based: every defect is cited to the reference file (and verified by direct read where the extractionâ€™s claim was ambiguous).
>
> **Decision vocabulary** - **ADOPT** â€” reuse mostly as-is (rename/port only; no logic change). Â§11.4.74 reuse. - **REFACTOR** â€” reuse the structure, fix bounded defects (1â€“N targeted edits). - **REWRITE** â€” defects too deep / design wrong; rebuild from the contract, may salvage shape. - **NET-NEW** â€” does not exist in the reference; must be authored.
>
> Anti-bluff (Â§11.4): a module is ADOPTed only when reading the source shows it is genuinely sound. Defective code is never ADOPTed just because it exists.

Reference path prefix (note the spaces), abbreviated `<REF>/` below: `docs/research/browser_extension/Browser Torrent Extension Guide/src/src/`

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1. Parser layer

Module/File	Quality assessment (defect cited or â€œsoundâ€�)	Decision	Required fixes	Risk
`parser/bencode.ts`	**Sound.** Zero-dep `Uint8Array` bencode; dict keys sorted lexicographically (`bencode.ts:139-151`) â€” load-bearing for stable infohash; rejects trailing data, unterminated ints/strings/lists/dicts; bounds-checks byte-strings; binary-vs-utf-8 mode; `sha1` via Web Crypto. Strongly unit-tested (`bencode.test.ts`).	**ADOPT**	Remove dead-assert in test (`bencode.test.ts:156`). Add property/fuzz + deep-recursion-limit (chaos) tests per Â§11.4.85. No source change.	Low
`parser/magnet.ts`	**Sound.** Repeated-key query parsing (multiple `tr`/`xt` preserved), hex+base32 infohash, `base32ToHex` RFC4648, full param extraction (xt/dn/tr/ws/xl/xs/kt/as/mt), recurse on base32. Fixture-driven tests.	**ADOPT**	Wire the unused 7-case `invalidMagnets` negative fixture. Consider BitTorrent v2 (`btmh`/multihash) â€” out of scope unless required.	Low
`parser/torrent-file.ts`	**Sound but DEAD CODE.** `parseTorrentFile`/ `computeInfohash` correct (infohash =	**REFACTOR**	Source is sound â†’ keep. The fix is **wiring**: invoke `computeInfohash` in the scanner/adapter to derive	Med (wiring touches scanner +

Module/File	Quality assessment (defect cited or â€œsoundâ€�)	Decision	Required fixes	Risk
	`sha1(encode(infoDict))`, BT v1 canonical) and tested â€” **but never invoked** by scanners/ adapter/background (sends upload raw `File`). Re-encode reorders non-sorted info keys (hash-divergence caveat, `torrent-file.ts:46-48`).		stable ids + local dedup against Boba. Â§11.4.124 (donâ€™t delete unwired code â€” wire it).	adapter contracts)

2. Scanner layer

Module/File	Quality assessment	Decision	Required fixes	Risk
`scanner/base.ts`	**One real defect.** Abstract base + shadow-DOM-aware `querySelectorAllDeep` + hidden-element skip + re-entrancy guard are sound. **Defect (verified by read):** `hashString` returns `Math.abs(hash).toString(36) + Date.now().toString(36)` (`base.ts:272`) â€” **time-salts the id** despite the doc-comment â€œstable IDâ€�, so the same torrent gets a new id every scan; cross-scan/cross-tab dedup is unreliable.	**REFACTOR**	Drop the `Date.now()` salt; derive id from `infohash` (now available once `torrent-file.ts` is wired) so ids are stable. One-method change.	Med (dedup correctnes depends o it)
`scanner/link-scanner.ts`	**Sound.** Site-specific pass (resolve selectors by domain) + generic `a[href]` fallback, dedup by normalized href, relativeâ†’absolute resolution, `sameOrigin` flag.	**ADOPT**	Reconcile the selector source (see `site-db` row).	Low
`scanner/text-scanner.ts`	**Sound.** `TreeWalker(SHOW_TEXT)` with skip-list + min-length filter, `findMagnetUris` dedup. Catches forum-pasted magnets.	**ADOPT**	None.	Low
`scanner/orchestrator.ts`	**Sound.** Coordinates scanners + MutationObserver, per-site debounce, cooperative `yieldToBrowser()` between scanners, relevant-mutation filter, emits scan events, dedups by id into a Map. Initial-state-only tested (orchestrator never actually scans in the reference suite).	**ADOPT**	Depends on the `base.ts` stable-id fix for reliable `mergeResults` dedup. Add real scan + mutation-rescan + debounce tests (the reference does not exercise scanning).	Med (dedup correctnes
`scanner/site-db.ts`	**Defect: two divergent selector tables.** `SITES` (15 rich entries, used ONLY for per-site debounce) overlaps and disagrees with `constants.SITE_SELECTORS` (~21 flat entries, used by `LinkScanner`). The private trackers	**REFACTOR**	Merge into ONE authoritative site registry (single source of truth) covering Bobaâ€™s real trackers (rutracker, kinozal,	Med (3-way reconcile: constants - site-db + SQL seed)

Module/File	Quality assessment	Decision	Required fixes	Risk
	Boba actually supports (rutracker/ iptorrents/kinozal/nnmclub) are split across both.		nnmclub, rutor, IPTorrents). Reconcile against the SQL `site_selectors` seed (30 rows) too.	

3. API layer

Module/File	Quality assessment	Decision	Required fixes	R
`api/client.ts`	**Sound HTTP client, WRONG TARGET.** TokenBucket rate-limit, AbortController timeout, `retryWithBackoff`, 429â†'`RateLimitError`, content-type branching â€" all sound. **But it speaks ONLY raw qBittorrent WebUI v2** (`/api/v2/...`) and the default base assumes qBittorrent on `:8080`. Nothing reaches Bobaâ€™s real services (7186 proxy / 7187 merge / 7189 jackett). `set-cookie` SID parse (`client.ts:128-166`) is browser-fragile.	**REFACTOR**	Keep the transport (rate-limit/retry/ timeout/error mapping). Re-point base URL/ endpoints to Bobaâ€™s real ports and routes; add a Boba-endpoint path set distinct from the raw qBt path set. Harden SID handling.	**H** (t re sa ex re ta is lo b in w
`api/auth.ts`	**Mostly sound, one crypto-coupling defect.** Four auth methods + failure backoff + `refreshIfNeeded` are sound. `createCredentialsFromConfig(config, passphrase)` (`auth.ts:170`) decrypts via `shared/crypto` â€" its correctness is hostage to the broken passphrase provenance (see crypto rows). Reads `client["authCookie"]` private field (`auth.ts:247`) â€" fragile TS escape hatch.	**REFACTOR**	Re-wire to the real key source (NET-NEW below). Replace the private-field read with a public accessor on the client. Decide which auth model Boba actually uses (delegate to proxy/ jackett vs raw qBt login).	H (c to cr re
`api/qbittorrent.ts`	**Sound.** Domain wrapper: `sendTorrent`/ `sendTorrents` (250ms spacing), `buildAddOptions` maps `ServerConfig`â†'add params, never rethrows (normalizes to `getUserMessage()`).	**ADOPT**	Inherits the re-targeted client. None of its own logic is defective.	L
`api/queue.ts`	**Defect: never authenticates (verified by read).** Enqueue/dequeue/FIFO-eviction/persist sound. But `processQueue` (`queue.ts:170`) builds `new BobaAPIClient(config.url,...)` + adapter and runs the send loop **with no** `login()`/ `authenticate()` **call** â€" retries against an authed server 403/fail-loop forever.	**REFACTOR**	Inject auth into `processQueue` (build+authenticate the client, or accept an authenticated client/adapter). Add retry/ attempts/backoff	M (c re si b w

Module/File	Quality assessment	Decision	Required fixes	R
			tests (reference only asserts structure).	
`api/health.ts`	**Sound, wrong port list.** Health thresholds, caching, `testConnection`, `autoDiscover`. **Defect:** `autoDiscover` probes `[8080, 7187, 7189]` (`health.ts:201`) â€” 8080 is wrong for Boba (qBt WebUI is proxied on 7186).	**REFACTOR**	Change probe port set to Bobaâ€™s real map (7186/7187/7189).	L

4. Content layer

Module/File	Quality assessment	Decision	Required fixes	Risk
`content/index.ts`	**Sound.** Wires orchestratorâ†’background (`scan-result`), message handler (`scan-now`/`get-detected`/`toggle-selection`/`get-scan-status`). **Defect (contract):** content `get-detected` returns `{torrents:[â€¦]}` while background `get-detected` returns `{data:{result}}` â€” divergent shapes the popup juggles.	**REFACTOR**	Unify the `get-detected` response contract across content + background + popup.	Med
`content/scanner.ts`	**Sound.** Thin content-side orchestrator glue (90 lines).	**ADOPT**	None.	Low
`content/highlight.ts`	**Sound.** Three highlight styles (badge/border/glow), event-bus driven.	**ADOPT**	None (presentation).	Low
`content/styles.css`	Presentation only, `.bobalink-*` + `!important`.	**ADOPT**	Rebrand class prefix if extension is renamed.	Low

5. Entrypoints (background / popup / options)

Module/File	Quality assessment	Decision	Required fixes	Risk
`background/index.ts`	**Sound hub, ONE critical defect.** Lifecycle, message router, context menus, commands, alarms (keepalive ~20s / health 5min), badge/notifications â€” all	**REFACTOR**	Replace the `""` decrypt call-site with the real key source. Fix `:8080` fallback. Add the missing background unit/	**HIGH** (credential correctness + the unrouted message types decision)

Module/File	Quality assessment	Decision	Required fixes	Risk
	sound. **Critical (verified `index.ts:409`):** `sendTorrents` decrypts creds with **empty passphrase** `""` â€" can only ever read `""`-encrypted bundles, and is mutually incompatible with the options pageâ€™s `"bobalink-extension"` encryption â‡’ the whole credential path is broken end-to-end. `open-dashboard` fallback hardcodes `:8080`. Has zero unit coverage in the reference.		integration coverage.	
`popup/popup.ts` + `popup/index.html` + `popup/styles.css`	**Sound.** Per-tab list, select/deselect, send, connection-status dot, empty-state scan. Excluded from reference coverage.	**ADOPT**	Adapt to the unified `get-detected` contract. Add real UI/UX behavior tests (reference e2e asserts presence only).	Low-Med
`options/options.ts` + `index.html` + `styles.css`	**CRITICAL defect (verified `options.ts:327`).** Formâ†”ï¸config bridge, server modal, test-connection, settings toggles all sound. **But every secret is AES-encrypted with the literal fixed passphrase** `"bobalink-extension"` â€" trivially reversible by anyone with the source (Â§11.4.10 leak). Auto-discover checkboxes are display-only (real	**REFACTOR**	Remove the fixed passphrase; route encryption through the real key source (NET-NEW). Wire the auto-discover checkboxes to actual config.	**HIGH** (Â§11.4.10 credential leak is a release blocker)

Module/File	Quality assessment	Decision	Required fixes	Risk
	ports hardcoded elsewhere).			

6. Shared libs

Module/File	Quality assessment	Decision	Required fixes	Risk
`shared/constants.ts`	**Sound values, WRONG PORTS + duplication.** Regexes, timeouts, retry/rate config, badge colors, encryption params all sound. **Defects:** `DEFAULT_PORTS.QBITTORRENT=8080` / `DEFAULT_URLS.QBITTORRENT` wrong (should be 7186); `SITE_SELECTORS` duplicates `site-db.SITES`; `RATE_LIMIT.MAX_REQUESTS=10` disagrees with DB seed `behavior.rate_limit_requests=30`.	**REFACTOR**	Fix qBt port 8080â†’7186 across host_permissions/CSP/URLs/PORTS/auto-discover. Collapse the duplicate selector table. Pick canonical rate-limit.	Med (port fix is cross-cutting: constants + manifest + SQL + auto-discover)
`shared/crypto.ts`	**SOUND PRIMITIVE â€" extractionâ€™s â€œREWRITEâ€� is too broad (DISAGREE, see Â§8).** Verified by read: PBKDF2 100k iters / AES-256-GCM / fresh 16-byte salt + 12-byte IV per op / 128-bit GCM tag / proper `StorageError` wrapping / rejects empty plaintext+passphrase. The crypto *module itself is correct and reusable*. The defect lives entirely in the **callersâ€™** passphrase (`options.ts:327`, `background:409`), not here.	**ADOPT** (module)	None to the module. The fix is the NET-NEW key source feeding it. Rewrite the **test** to import this real module (reference test tests a copy).	Low (module) / High (caller key source â€" separate NET-NEW)
`shared/errors.ts`	**Sound.** Typed error taxonomy (`BobaLinkError` + 8 subclasses), `getUserMessage`, `normalizeError`, `isBobaLinkError`.	**ADOPT**	Add direct error-class unit tests.	Low
`shared/events.ts`	**Sound.** `TypedEventEmitter` + 13-event `EventMap`, per-listener try/catch isolation, unsub returns. Thoroughly tested.	**ADOPT**	None.	Low
`shared/logger.ts`	**Sound.** Structured leveled logging, `timed`, `createLogger`, debug-gating.	**ADOPT**	None.	Low
`shared/storage.ts`	**Sound.** `chrome.storage.local` wrapper, `NamespacedStorage`, change listener, `bobalink_`-prefixed clear. **Note:** only the JSON-blob model is implemented here â€" the parallel SQL-schema (sql.js) model has NO wrapper file.	**ADOPT**	Decide one persistence source of truth (JSON-blob vs sql.js); the sql.js wrapper is NET-NEW if chosen. Add direct storage unit tests (always mocked away in the reference).	Med (dual-storage architecture decision)

Module/File	Quality assessment	Decision	Required fixes	Risk
`shared/utils.ts`	**Sound.** debounce/throttle, `retryWithBackoff`, `TokenBucket`, `processInChunks`, `formatBytes`, url helpers. **Note:** `escapeHtml` is DOM-dependent (not service-worker safe); `processInChunks` defined but unused.	**ADOPT**	Provide a SW-safe `escapeHtml` variant for background context.	Low

7. Types

Module/File	Quality assessment	Decision	Required fixes	Risk
`types/torrent.ts`	**Sound.** `MagnetInfo`/`TorrentFile`/`ParsedTorrent`/`DetectedTorrent`/`SendResult`/`PageScanResult`.	**ADOPT**	None.	Low
`types/config.ts`	**Sound, with duplication.** `ServerConfig`/`ExtensionConfig`/`DEFAULT_CONFIG`. **Defect:** `AuthMethod` duplicated in `types/api.ts`; `ServerConfig` (TS) diverges from `server_config` (SQL) field naming.	**REFACTOR**	De-duplicate `AuthMethod` (single canonical). Reconcile TS `ServerConfig` â†”ï¸Ž SQL `server_config`. Fix default ports (auto-discovery `[7187,7189,8080]`â†’ Boba map).	Low
`types/api.ts`	**Sound, declares unused contracts.** qBt/auth/health/queue/message types. `BobaServerInfo`/`BobaSearchResponse` declared but **nothing consumes them** (no Boba-API client). `MessageType` declares 7 values with no handler branch.	**REFACTOR**	De-dup `AuthMethod`. Decide which declared-but-unrouted message types to implement vs drop. The Boba-API consumer that uses `BobaSearchResponse` is NET-NEW.	Low (types) / High (the consumer is net-new)

8. SQL schema

Module/File	Quality assessment	Decision	Required fixes	Risk
`sql/schemas.sql` (=`migrations/001_initial.sql`)	**Sound schema, wrong default + unresolved dual-storage.** 9 tables, ~25 indexes, partial indexes, encrypted-credential columns (`api_key_encrypted/`	**REFACTOR** (if SQL chosen) / **drop** (if JSON-blob chosen)	First decide the single persistence model. If SQL: fix port seeds	Med (architecture decision gates everything)

Module/File	Quality assessment	Decision	Required fixes	Risk
	`password_encrypted`), 30 selector seeds, 25 config seeds â€” well-formed SQLite 3.38+/sql.js. **Defects:** seeds `qbittorrent_port DEFAULT 8080` + `server.base_url=http://localhost:8080`; and the runtime has NO sql.js wrapper (only the JSON-blob storage path is implemented) so it is currently unused-by-design.		8080â†'7186, reconcile site-selector seed with the merged site registry, author the sql.js wrapper (NET-NEW). If JSON-blob: this schema is not adopted.	

9. Build / test configs

Module/File	Quality assessment	Decision	Required fixes	Ri
`wxt.config.ts`	**Sound MV3 manifest gen, wrong endpoints.** WXT manifest, 6 permissions, CSP, commands, auto-icons. **Defects:** host_permissions + CSP `connect-src` include `:8080` (should be `:7186`); icon PNGs referenced but absent.	**REFACTOR**	Fix 8080â†'7186 in host_permissions + CSP. Confirm Firefox MV3 module-SW support.	Me
`tsconfig.json`	**Sound.** Very strict (all strict flags + `noUncheckedIndexedAccess` + `exactOptionalPropertyTypes`).	**ADOPT**	None.	Lo
`package.json`	**Sound, drift.** Deps/scripts fine. **Defects:** HTTPS repo URL (Boba mandates SSH); release.yml references non-existent `build:chrome` script.	**REFACTOR**	SSH remote; rebrand name; align scripts; drop CI-only assumptions.	Lo
`jest.config.ts`	**Defect: 80% threshold + UI/entrypoints excluded from coverage.** `collectCoverageFrom` excludes `popup/options/content/background/**/index.ts` â€” UI/entrypoint logic has ZERO coverage enforcement. Conflicts with Â§11.4.27.	**REWRITE** (or replace w/ Vitest)	Decide runner (Vitest recommended â€” matches Boba `frontend/` + the crypto test). Bring UI/entrypoints INTO coverage; raise toward 100%.	Me dec fou
`playwright.config.ts`	**Defect: non-functional e2e config.** References missing `global-setup.ts`/`global-teardown.ts`; dead `webServer.url :3000`; Firefox project canâ€™t load extension via Chromium args;	**REWRITE**	Author global-setup/teardown (build `dist`, launch persistent Chromium with `--load-`	Me

Module/File	Quality assessment	Decision	Required fixes	Ris
	`reporter:github`+`forbidOnly:CI` assume GitHub Actions.		`extension`, capture real extension id, rewrite `chrome-extension://<id>/...`). Remove GH-Actions assumptions (Boba manual CI).	
`.eslintrc.json` / `.prettierrc`	**Sound.** Type-aware lint (`no-explicit-any`/`no-floating-promises` errors), Prettier 100-col.	**ADOPT**	Align with Boba project lint config.	Lo
`_locales/en/messages.json`	**Sound.** 28 i18n keys.	**ADOPT**	Rebrand strings if renamed.	Lo
`.github/workflows/ci.yml`, `release.yml`	**FORBIDDEN.** Boba Hard-Stop: no `.github/workflows/*.yml` ever.	**drop (do NOT port)**	Extract only the *commands* (lint/compile/test/build/zip/sha256) into a manual `ci.sh`-style script. Never copy the YAML.	Lo con har

10. Tests

Module/File	Quality assessment	Decision	Required fixes	Risk
`bencode.test.ts`	Sound; dead-assert line 156.	**ADOPT**	Remove dead assert; add fuzz/chaos.	Low
`magnet.test.ts`	Sound, fixture-driven; `invalidMagnets` (7) unused.	**ADOPT**	Wire negative fixtures.	Low
`torrent-file.test.ts`	Sound (infohash determinism, single/multi parse).	**ADOPT**	None.	Low
`api-client.test.ts`	**Partial BLUFF.** Auth block ends `expect(true).toBe(true)` (no observable check); 429 test asserts `ServerError` despite name saying `RateLimitError`.	**REFACTOR**	Replace the bluff Auth assertions with real cookie-header propagation checks; fix the 429 assertion.	Med
`queue.test.ts`	**Partial BLUFF.** Auto-processing block ends `expect(true).toBe(true)`; processQueue only asserts *structure*, not success/failure.	**REFACTOR**	Assert real retry/attempts/outcomes; remove no-ops.	Med

Module/File	Quality assessment	Decision	Required fixes	Risk
`scanner.test.ts`	**Weak.** Event-emitter block sound; but orchestrator block asserts initial-state only (never scans); DOM block tests raw `querySelectorAll`, NOT the production scanner.	**REFACTOR**	Drive the real orchestrator over real DOM; assert detected torrents + mutation rescans.	Med
`crypto.test.ts`	**BLUFF (Â§11.4.1).** 44 thorough tests â€” but against an **inline re-implemented copy**, never `src/shared/crypto.ts`. Cannot fail against a no-op stub of the real module. Also imports `vitest` while the other 6 files use Jest (mixed runners).	**REWRITE**	Re-point every test at the production `shared/crypto` module; standardize runner.	Med (itâ€™s the most-thorough file but proves nothing as-is)
`chrome-mock.ts` / `setup.ts`	Mock infra; happy-path only (no error injection), no `cookies`/`scripting`/`sync`; `setup.ts:14` wrong path (type-only).	**REFACTOR**	Add error-injecting variant for chaos/stress; fix the stale path.	Low
`content.spec.ts` / `options.spec.ts` / `popup.spec.ts` (e2e)	**NON-FUNCTIONAL.** Hardcoded `chrome-extension://test-id/...`; no fixture loads the extension or resolves the real id; content spec asserts raw DOM, not the injected content script. Asserts presence/visibility, never user outcomes.	**REWRITE**	Real extension-loading fixture + user-observable assertions (torrent rendered, server persisted, sendâ†’appears in qBittorrent).	High (load-bearing for Â§11.4 anti-bluff e2e)

11. NET-NEW (does not exist in the reference â€” must be authored)

Item	Why net-new	Risk
Real key source for credential encryption	crypto.ts is a sound primitive but has NO real passphrase provenance â€" callers use a fixed string + an empty string. Need a master-passphrase prompt, OS-keychain, OR delegation to boba-jackett/ proxy (which already own encrypted creds at `/config/boba.db`). Â§11.4.10 blocker.	**HIGH**
Real Boba service integration (7186/7187/7189)	Reference speaks ONLY raw qBittorrent v2; `BobaServerInfo`/ `BobaSearchResponse` types are declared but nothing consumes them. The merge-search API (7187), tracker-cookie download path (7186 proxy), and boba-jackett (7189) clients do not exist.	**HIGH**
`chrome.tabGroups` **tab-group batching**	Verified: the reference has NO `chrome.tabGroups` usage. The only â€œbatchingâ€� is intra-tab `yieldToBrowser()` cooperative yielding + per-tab `tabTorrents` map + an unused `processInChunks`. Tab-group batch behavior is fresh design.	Med
Icon PNG rasterization (16/32/48/128)	Verified: `src/assets/` contains `icon.svg` only â€" zero `.png` files. Code/manifest reference rasterized PNGs that must be produced at build (WXT auto-icons from the SVG).	Low
sql.js storage wrapper (only if SQL persistence chosen)	The 9-table schema exists but NO runtime wrapper implements it; only the JSON-blob `storage.ts` path is live.	Med
Playwright `global-setup.ts` / `global-teardown.ts`	Referenced by config, absent from tree. Must build `dist`, launch persistent Chromium `--load-extension`, capture real extension id.	Med

Item	Why net-new	Risk
Missing test TYPES (Â§11.4.27/Â§11.4.85): integration (real cross-module â†’ live 7186/7187), functional e2e, automation/HelixQA, security/pen (credential-leak, XSS via magnet `dn`, CSP, permission-scope), DDoS/load, scaling, chaos (storage corruption / network fault / SW termination), stress, performance, benchmark, UI/UX, Challenges	Reference covers only unit + e2e (and those have bluffs). 13 of 15 mandated types absent.	**HIGH** (volume)

Reuse estimate

Counting the **44 `.ts` source files** plus the SQL schema and the build/test configs:

Disposition	Approx. count	Examples
ADOPT (reuse ~as-is)	~17	bencode, magnet, link/text-scanner, content/scanner+highlight, qbittorrent adapter, crypto module, errors, events, logger, storage, utils, types/torrent, eslint/prettier/tsconfig, i18n
REFACTOR (bounded fixes)	~18	torrent-file (wire), base (un-salt id), site-db (merge tables), client (re-target), auth, queue (auth), health (ports), content/index (contract), background, options, popup, constants (ports), types/config, types/api, wxt.config, package.json, chrome-mock, sql schema
REWRITE	~7	jest.config (runner + coverage), playwright.config, crypto.test, api-client.test (bluff), queue.test (bluff), scanner.test, all 3 e2e specs
NET-NEW	7+ areas	key source, Boba-service clients, tab-groups, icon PNGs, sql.js wrapper, e2e global setup, 13 missing test types
DROP	2	both `.github/workflows/*.yml` (constitution hard-stop)

Headline: roughly 80% of the referenceâ€™s logic/structure is reusable (ADOPT + REFACTOR â‰ˆ 35 of ~44 source-bearing files â€” the parsers, scanners, shared libs, types, and the API transport are sound), **~20% must be rewritten** (the test suiteâ€™s bluffs/non-functional e2e + the runner/coverage configs), **and the highest-value work is NET-NEW integration** (real Boba endpoints + a real key source) that the reference never attempted. The reference is a strong *skeleton with sound primitives* whose two structural gaps â€” credential key-management and Boba-service integration â€” are exactly the parts it left as placeholders.

Top 5 highest-risk rewrites

1. **Real key source for credential encryption (NET-NEW).** The sound crypto primitive is currently fed a fixed (`"bobalink-extension"`, `options.ts:327`) and an empty (`""`, `background:409`) passphrase that are even mutually incompatible â€” the credential path is broken end-to-end and is a Â§11.4.10 leak (release blocker). Touches options + background + auth.
2. **Boba-service integration / `api/client.ts` re-target (REFACTORâ†’effectively rewrite of the endpoint layer).** The client speaks only raw qBittorrent v2 on `:8080`; nothing reaches 7186/7187/7189. Re-targeting + adding the Boba-API consumer (`BobaSearchResponse`) is the load-bearing integration, and the SID/cookie + CORS/host-permission story is fragile.
3. **Functional e2e suite (REWRITE all 3 specs + author global-setup/teardown).** Every reference e2e is non-functional (hardcoded `chrome-extension://test-id`, no extension load, presence-only asserts). Â§11.4 anti-bluff demands real extension load + user-observable outcomes â€” this is from-scratch.
4. `crypto.test.ts` **(REWRITE).** The single most thorough test file (44 cases) is a Â§11.4.1 bluff â€” it tests an inline copy, never the production module, so it passes against a no-op stub. Must be re-pointed at `shared/crypto` and the runner standardized (mixed Jest/Vitest).
5. **Test runner + coverage config (jest.config REWRITE).** 80% threshold with UI/entrypoints excluded conflicts with Â§11.4.27; the runner decision (Vitest vs Jest) is foundational and gates every other test fileâ€™s port. (Close behind: `queue.ts` auth-gap fix + the `base.ts` time-salted-id fix, both correctness-critical for offline retries and dedup.)

Disagreements with the extractionâ€™s defect claims (re-verified by reading source)

1. `crypto.ts` **â†’ the extraction tags it â€œðŸ”´ REWRITEâ€�; I DISAGREE â€” the MODULE is ADOPT, the KEY SOURCE is NET-NEW.** Direct read of `shared/crypto.ts` (lines 42-228) shows the encrypt/decrypt machinery is genuinely sound: PBKDF2 100k iterations, AES-256-GCM, a fresh 16-byte random salt + 12-byte IV per operation, a 128-bit GCM auth tag, and correct `StorageError` wrapping that rejects empty plaintext/passphrase. **No defect exists inside this file.** The defects the extraction cites â€” fixed passphrase `"bobalink-extension"` (verified `options.ts:327`) and empty passphrase `""` (verified `background/index.ts:409`) â€” live entirely in the *callers*, not in the crypto module. So the correct disposition is: **ADOPT the crypto primitive unchanged; author the real key source as NET-NEW; REFACTOR the two call-sites.** Rewriting the crypto module itself would be discarding sound, reusable work (Â§11.4.74). The extractionâ€™s â€œðŸ”´ crypto â€¦ â†’ REWRITEâ€� headline conflates the module with its misuse.

2. **Minor: the extraction calls `base.ts` id generation a generic â€œunstable / time-saltedâ€� concern; verified it is a one-line REFACTOR, not a rewrite.** `base.ts:272`

literally appends `Date.now().toString(36)` to the hash; removing that suffix and sourcing the id from the (now-wireable) infohash is a bounded fix to a single private method â€" the rest of `base.ts` (shadow-DOM traversal, hidden-element filtering, re-entrancy guard) is sound and ADOPT-grade.

All other extraction defect claims I sampled were verified accurate: the queue never authenticates (`queue.ts:170` builds+runs with no login), the `.torrent` parser is genuinely never invoked, the two selector tables genuinely diverge, the icon PNGs are genuinely absent (only `icon.svg`), and the two passphrase call-sites are genuinely incompatible.